



## Technical Reproducibility Validation Report

### HipSTR

Version 12e989b

Autosomal STR

|                   |                                                                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Verification date | 2026-09-20                                                                                                                              |
| Repository commit | <a href="https://github.com/STRhub/HipSTR/commit/12e989be4a8f9ab59f0c4c5da3784b82018cac82">12e989be4a8f9ab59f0c4c5da3784b82018cac82</a> |
| Result            | <b>Runs as documented</b>                                                                                                               |
| Reached           | Runs + Plausible output                                                                                                                 |
| Permalink         | <a href="https://strhub.app/verified/hipstr">https://strhub.app/verified/hipstr</a>                                                     |
| STRhub            | <a href="https://strhub.app">https://strhub.app</a>                                                                                     |

### SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

#### WHAT WAS VERIFIED

- Source available
- Installation successful
- End-to-end execution
- Output generated

#### NOT VERIFIED

- Genotype accuracy
- Concordance
- Forensic validity
- Regulatory compliance



## 1. Executive Summary

|               |                                                                                         |
|---------------|-----------------------------------------------------------------------------------------|
| PURPOSE       | Verify that the tool installs, runs end-to-end, and produces structurally valid output. |
| RESULT        | <b>Runs as documented</b>                                                               |
| REACHED       | Runs + Plausible output, 5/5 gates passed                                               |
| WHY           | The tool installed and its run produced its documented output.                          |
| SCOPE         | Installation, execution, and output structure verification.                             |
| NOT EVALUATED | Genotype accuracy · Concordance · Forensic validity · Regulatory compliance             |

## 2. As it is in the repository: Verification Gates

|                  |                                                           |      |
|------------------|-----------------------------------------------------------|------|
| Available        | The pinned public source exists                           | PASS |
| Installs         | The environment builds from source                        | PASS |
| Runs             | Executes end-to-end without crashing                      | PASS |
| Expected IO      | Produces a non-empty file in the declared format          | PASS |
| Plausible Output | Output structure matches expected genotype-bearing format | PASS |
| Execution log    | <a href="#">hipstr.log-external.txt</a>                   |      |

## 3. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · HipSTR 12e989b · commit 12e989b
$ cd '/opt/tool' &&
touch /tmp/.strhub_mark &&
( ./HipSTR \
--bams /data/in/input.bam \
--fasta /data/ref/hg38.fa \
--regions /data/in/regions.bed \
--str-vcf str_calls.vcf.gz \
); rc=$?; find . \
-type f \
-newer /tmp/.strhub_mark \
-size +0 \
! \
-path './.git/*' \
2>/dev/null |
head \
-500 |
while IFS= read \
-r f; \
do mkdir \
-p "/data/out/${dirname \
"$f")" &&
cp "$f" "/data/out/$f"; done; exit $rc
```

## 4. Reproducibility Metadata

|                 |                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tool            | HipSTR                                                                                                                                                |
| Version         | 12e989b                                                                                                                                               |
| Repository      | <a href="https://github.com/tfwillems/HipSTR">https://github.com/tfwillems/HipSTR</a>                                                                 |
| Commit (pinned) | <a href="https://github.com/tfwillems/HipSTR/commit/12e989be4a8f9ab59f0c4c5da3784b82018cac82">12e989be4a8f9ab59f0c4c5da3784b82018cac82</a>            |
| Container OS    | ubuntu-22.04                                                                                                                                          |
| Marker panel    | Autosomal STR                                                                                                                                         |
| Verified on     | 2026-09-20 12:53:45 UTC                                                                                                                               |
| CI run          | <a href="https://github.com/Tfronta/strhub-verified/actions/runs/35511745510">https://github.com/Tfronta/strhub-verified/actions/runs/35511745510</a> |

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

## 5. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

## 6. Output Content Evidence

|                          |                  |
|--------------------------|------------------|
| Output file              | str_calls.vcf.gz |
| Format                   | VCF              |
| Sequence records         | 21               |
| Distinct STR loci        | 21               |
| Total reads              | 7,851            |
| Max depth (single locus) | 517              |

## 7. Verification Data

The repository ships example data (3 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 24 forensic STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

|             |                                                                                                                                                                                              |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset     | Illumina BAM (hg38), NA12878 (autosomal, female)                                                                                                                                             |
| Source      | <a href="https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat...">https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat...</a>                                                |
| License     | Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.                                                                                                    |
| Ref. genome | GRCh38 / hg38                                                                                                                                                                                |
| Loci tested | CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D18S51 · D19S433 · D1S1656 · D22S1045 · D2S441 · D3S1358 · D5S818 · D6S1043 · D7S820 · D8S1179 · FGA · PentaD · PentaE · TH01 · TPOX · vWA |
| Regions BED | Provided by STRhub                                                                                                                                                                           |

## 8. Verification Matrix

|             |                             |                                                  |
|-------------|-----------------------------|--------------------------------------------------|
| <b>PASS</b> | Reference dataset           | Illumina BAM (hg38), NA12878 (autosomal, female) |
|             | README check (5/5)          | Advisory: all items present                      |
|             | Install / environment setup | Present                                          |
|             | Run command                 | Present                                          |
|             | Expected input format       | Present                                          |
|             | Produced output format      | Present                                          |
|             | Dependencies / versions     | Present                                          |

## 9. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

- Test data: no sample from the repository was used, so a public reference sample stood in.
- A container environment, built from the tool's declared install steps rather than from a recipe the repository ships.

## 10. Notes from Reading the Repository

Recorded automatically from the tool's public files when this run was configured. Not verified by execution, and not part of the gates above. Useful for what to check by hand.

- Environment: generated by STRhub from the repository's Makefile (make), at the pinned commit.
- Input: the README documents BAM (line 69); this run used STRhub's illumina-bam-hg38 reference data.
- 1 input path(s) in the README command replaced with /data/in/input.bam.
- 1 reference FASTA path(s) replaced with /data/ref/hg38.fa.
- 1 BED path(s) replaced with /data/in/regions.bed.
- Run command: the README's own command, rewritten to STRhub's mounts; everything it created was captured as output.
- Regions: STRhub's ready-made hipstr file for the dataset's panel loci (STRhub records the hipstr layout for a program of this name).

## 11. Limitations

- Single reference dataset per input type
- Single containerized environment (Docker / ubuntu-22.04)
- No truth-set comparison or ground-truth genotypes
- No accuracy or concordance assessment
- No forensic validation of results
- Short-read limitations apply (very long STR alleles may not span reads)

## 12. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

## 13. Conclusion

### ◆ Runs end-to-end

HipSTR 12e989b installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit.

### ◆ Produces plausible output

The tool generated a structurally valid VCF output file with genotype calls across 21 target forensic STR loci, with a maximum read depth of 517 at D18S51.

### ◆ Verdict: Runs

The tool installed and its run produced its documented output.

### ◆ Repository test data not used

The repository ships example data (3 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878 (autosomal, female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

### ◆ Reproducibility statement

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.



## Appendix: Detected Markers with Read Depth

HipSTR 12e989b · verified 2026-09-20

| LOCUS    | READ DEPTH |                      | LOCUS    | READ DEPTH |                  |
|----------|------------|----------------------|----------|------------|------------------|
| D18S51   | 517        | ████████████████████ | D6S1043  | 360        | ████████████████ |
| FGA      | 477        | ██████████████████   | D1S1656  | 359        | ████████████████ |
| vWA      | 456        | ██████████████████   | D12S391  | 349        | ██████████████   |
| PentaD   | 431        | ████████████████     | D8S1179  | 346        | ██████████████   |
| D2S441   | 428        | ████████████████     | D13S317  | 316        | ██████████       |
| D3S1358  | 419        | ████████████████     | D10S1248 | 309        | ██████████       |
| D22S1045 | 414        | ████████████████     | D19S433  | 289        | ████████         |
| D7S820   | 405        | ████████████████     | TH01     | 275        | ████████         |
| PentaE   | 401        | ████████████████     | CSF1PO   | 275        | ████████         |
| D5S818   | 399        | ████████████████     | TPOX     | 255        | ██████           |
| D16S539  | 371        | ██████████           |          |            |                  |